

4-Tier Architecture Overview

- Each tier has specific responsibilities, and separating concerns across more layers allows for greater scalability, maintainability, and flexibility in larger systems.
- Example of 4-Tier Architecture in an E-Commerce System

- **Presentation (Client Tier):**

- The user interface where customers interact with the e-commerce platform. This could be a web application or mobile app where customers browse products, add to cart, and check out.

- **Business Logic (Application Tier):**

- This layer contains the business logic, which handles core operations like processing orders, applying discounts, calculating shipping costs, and managing customer sessions.

- **Data (Persistence Tier):**

- This tier manages the storage of data such as customer information, product catalogs, order details, payment history, etc. It is typically managed by a database.

- **Integration (Service Tier):**

- The integration layer connects the e-commerce system with third-party services like payment gateways, shipping providers, external marketing tools, CRM systems, inventory management systems, etc. This is essential for ensuring that the e-commerce platform can interact with external services efficiently and securely.

Client Tier

- where the customer interacts with the e-commerce system. It is responsible for displaying information to users and collecting their inputs. This layer can include a website, mobile apps, or other front-end interfaces.
- **Components:**
- **Web Interface:** HTML, CSS, JavaScript, React, Angular, or Vue.js to render the UI.
- **Mobile App:** Built with Swift (iOS) or Kotlin/Java (Android).
- **User Actions:**
 - Browsing products.
 - Adding items to the cart.
 - Filling in billing and shipping information.
- Checking out and making a payment. **Example:**
- A customer visits the website (e.g., www.example-ecommerce.com) and sees the product listings, browses categories, selects an item, and adds it to their cart.

Business Logic (Application Tier)

- This layer holds the core functionality of the e-commerce system, where all the business rules and logic are applied. It processes requests from the presentation layer, communicates with the data layer to fetch necessary data, and handles complex transactions like order processing, promotions, and payment validation.
- **Components:**
- **Order Processing Logic:** Handle adding items to the cart, calculating totals, applying discounts, processing refunds, etc.
- **User Authentication & Authorization:** Managing user sessions, login, and role-based access control.
- **Business Rules:** Managing promotions, loyalty programs, personalized recommendations.
- **Payment Gateway Integration:** Verifying payment details and coordinating with payment providers.
- **Example:**
- When a customer clicks “Proceed to Checkout,” the system verifies the availability of the products in the cart, applies any discounts or promotions, and calculates the total price (including tax and shipping).

Business Logic (Application Tier)

- This layer holds the core functionality of the e-commerce system, where all the business rules and logic are applied. It processes requests from the presentation layer, communicates with the data layer to fetch necessary data, and handles complex transactions like order processing, promotions, and payment validation.
- **Components:**
- **Order Processing Logic:** Handle adding items to the cart, calculating totals, applying discounts, processing refunds, etc.
- **User Authentication & Authorization:** Managing user sessions, login, and role-based access control.
- **Business Rules:** Managing promotions, loyalty programs, personalized recommendations.
- **Payment Gateway Integration:** Verifying payment details and coordinating with payment providers.
- **Example:**
- When a customer clicks “Proceed to Checkout,” the system verifies the availability of the products in the cart, applies any discounts or promotions, and calculates the total price (including tax and shipping).

Data (Persistence Tier)

- This tier is responsible for storing and retrieving the data required by the e-commerce system. It typically involves relational databases, NoSQL databases, or even cloud storage, depending on the specific needs of the system.
- **Components:**
- **Database Management System (DBMS):** Stores critical data like user profiles, orders, products, payment information, inventory, and shipping details.
- **Data Models:** Tables or collections in the database for products, users, orders, payments, reviews, etc.
- **Technologies:** SQL databases (MySQL, PostgreSQL), NoSQL databases (MongoDB, DynamoDB), cloud-based databases (AWS RDS, Google Cloud SQL).
- **Example:**
- The **Products table** in the database stores product details (name, price, description, image, and stock quantity), and the **Orders table** stores customer purchase details (order ID, product ID, customer ID, shipping address, etc.).

Integration (Service Tier)

- The integration layer handles communication between the e-commerce platform and third-party services that are required for extended functionality. This could include services for payment processing, shipping, external APIs for marketing and CRM systems, or integration with external inventory management systems.
- **Components:**
 - **Payment Gateways:** Integrates with third-party services like **Stripe**, **PayPal**, or **Square** to handle secure payment processing.
 - **Shipping APIs:** Interfaces with services like **UPS**, **FedEx**, or **DHL** to handle shipment tracking and shipping rate calculations.
 - **Inventory Systems:** Communicates with external inventory management systems to ensure that stock levels are up to date and accurate.
 - **CRM Systems:** Integration with customer relationship management systems (e.g., **Salesforce**, **HubSpot**) for personalized marketing campaigns.
 - **Email Services:** Interfaces with services like **Mailchimp** or **SendGrid** to send order confirmations, promotional emails, or customer newsletters.
- **Example:**
 - When the customer enters their payment information and completes the checkout process, the system communicates with the **payment gateway** to process the payment. Simultaneously, it sends shipping details to the **shipping API** to generate a shipping label and track delivery.

How These tiers Work Together

1. Customer Interaction

The user browses products, adds items to the cart, and proceeds to checkout.

2. Business Logic Processing :

The system calculates taxes, shipping, and applies discounts. It also processes the payment through an external payment gateway and verifies user credentials.

3. Data Retrieval:

The business logic layer queries the database to check product availability, user information, payment history, and order details.

4. External Services:

The system interacts with third-party services like payment gateways and shipping APIs. After payment is processed successfully, the order information is sent to the shipping service for delivery.